

Architecture Before the Formula

Technical Supplement: Proofs, Variants, and Stochastic Extensions

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Abstract

This supplement collects quantitative bounds, edge cases, and extensions outside the main dependency spine of the v1.0 article. It includes entropy-regularized anchoring, softmax stability and score-rank approximation, structural approximation ledgers, finite-support interpolation, explicit witnesses separating the attention commitments, information accounting under aggregation, signed and normalized FFN presentations, typed local approximation, residual error propagation, and stochastic variable-successor realizations with conditional Gibbs and KL identities. None of these extensions changes the main article’s architectural primitive or the dependency path to the Transformer normal form.

Relation to the main article. The main article contains the definitions and proofs required for receiver-factored architecture, interface barriers, attention, local continuation, and the canonical block. This supplement assumes that notation and compiles after the main article so cross-document references can be resolved.

1 Additional proofs and quantitative results

The main article derives the exact receiver-interface, attention, aggregation, feedforward, and whole-block results. This section adds quantitative statements that are useful when one of the exact commitments is only approximately realized.

1.1 Entropy-regularized receiver-row anchoring

The main article derives normalized positive allocation from ratio preservation and unit receiver mass. The same row anchor is the unique solution of a receiver-local regularized optimization problem.

Proposition 1.1 (Entropy-regularized row anchor). *Let S be finite, let $\Gamma_i > 0$, let $s_i \in \mathbb{R}$, and let $\tau > 0$. With the convention $0 \log 0 = 0$, the problem*

$$\max_{p \in \Delta(S)} \left\{ \sum_{i \in S} p_i s_i - \tau \sum_{i \in S} p_i \log \frac{p_i}{\Gamma_i} \right\}$$

has the unique solution

$$p_i = \frac{\Gamma_i e^{s_i/\tau}}{\sum_{j \in S} \Gamma_j e^{s_j/\tau}}.$$

A hard mask is implemented by replacing S with the nonempty admissible support.

Proof. The objective is continuous on the compact simplex and therefore attains a maximum. Its entropy term is strictly concave on the relative interior, while the linear score term does not change concavity. Moreover, the one-sided derivative of $-p_i \log p_i$ diverges to $+\infty$ as $p_i \downarrow 0$, so the maximizer has positive coordinates on the declared support.

Introduce a multiplier λ for $\sum_i p_i = 1$. The first-order condition is

$$s_i - \tau \left(\log \frac{p_i}{\Gamma_i} + 1 \right) - \lambda = 0.$$

Hence

$$p_i = \Gamma_i \exp \left(\frac{s_i - \lambda - \tau}{\tau} \right) = c \Gamma_i e^{s_i/\tau}$$

for a common positive constant c . Normalization gives

$$c^{-1} = \sum_j \Gamma_j e^{s_j/\tau}.$$

Strict concavity gives uniqueness. □

The result supplies a second interpretation of row-softmax: it is not merely a normalization convention but the optimizer of a declared score-versus-relative-entropy tradeoff. That interpretation still depends on choosing a receiver-local simplex, a prior Γ , and an entropy regularizer. Other anchors correspond to different feasible sets or regularizers.

1.2 Softmax stability

For $z \in \mathbb{R}^n$ and temperature $\tau > 0$, write

$$\sigma_\tau(z)_i := \frac{e^{z_i/\tau}}{\sum_j e^{z_j/\tau}}.$$

Lemma 1.2 (Softmax Lipschitz bound). *For all $z, z' \in \mathbb{R}^n$,*

$$\|\sigma_\tau(z) - \sigma_\tau(z')\|_1 \leq \frac{2}{\tau} \|z - z'\|_\infty.$$

The same bound holds when both rows use the same fixed nonempty mask, after restricting to the admissible coordinates.

Proof. Let $p = \sigma_\tau(z)$. The Jacobian is

$$D\sigma_\tau(z) = \tau^{-1} (\text{Diag}(p) - pp^\top).$$

For any u with $\|u\|_\infty \leq 1$, set $\mu = p^\top u$. Then

$$\begin{aligned} \|(\text{Diag}(p) - pp^\top) u\|_1 &= \sum_i p_i |u_i - \mu| \\ &\leq 2 \sum_i p_i = 2, \end{aligned}$$

because $u_i, \mu \in [-1, 1]$. Thus the operator norm from ℓ_∞ to ℓ_1 is at most $2/\tau$. Integrating the Jacobian along the segment $z_t = z' + t(z - z')$ gives

$$\sigma_\tau(z) - \sigma_\tau(z') = \int_0^1 D\sigma_\tau(z_t)(z - z') dt,$$

and the stated bound follows. □

The bound makes temperature an explicit stability commitment. Smaller τ permits sharper allocations but amplifies score perturbations in this worst-case estimate.

1.3 Rank-restricted score approximation

Let carrier states satisfy $\|h_i\| \leq R$. Consider a bilinear score

$$s_{ij} = h_i^\top M h_j$$

and a rank- d approximation M_d .

Proposition 1.3 (Rank-truncated attention error). *Let p_i and $\hat{p}_i$ be the receiver-row softmax distributions generated by M and M_d at temperature τ on the same fixed nonempty support. Then*

$$\|p_i - \hat{p}_i\|_1 \leq \frac{2R^2}{\tau} \|M - M_d\|_{\text{op}}.$$

If $\|Vh_j\| \leq B$ and O is a linear output map, then

$$\left\| O \sum_j (p_{ij} - \hat{p}_{ij}) Vh_j \right\| \leq \frac{2\|O\|_{\text{op}} B R^2}{\tau} \|M - M_d\|_{\text{op}}.$$

For a best rank- d approximation in operator norm,

$$\|M - M_d\|_{\text{op}} = \sigma_{d+1}(M).$$

Proof. For every admissible pair,

$$|h_i^\top (M - M_d) h_j| \leq \|h_i\| \|M - M_d\|_{\text{op}} \|h_j\| \leq R^2 \|M - M_d\|_{\text{op}}.$$

Therefore the row-logit perturbation has ℓ_∞ norm at most $R^2 \|M - M_d\|_{\text{op}}$. Apply Lemma 1.2.

For the output,

$$\begin{aligned} \left\| O \sum_j (p_{ij} - \hat{p}_{ij}) Vh_j \right\| &\leq \|O\|_{\text{op}} \sum_j |p_{ij} - \hat{p}_{ij}| \|Vh_j\| \\ &\leq \|O\|_{\text{op}} B \|p_i - \hat{p}_i\|_1. \end{aligned}$$

Substitute the first bound. The final equality is the Eckart–Young characterization of the best rank- d operator-norm approximation. $\square$

A rank budget is therefore a quantitative commitment distinct from the structural claim that the score family is shared and separable. The latter supplies common query–key coordinates; the former controls the approximation defect of a restricted coordinate dimension.

1.4 A routed–local approximation ledger

The main article proves an exact decomposition for an instantiated canonical block. A separate question asks when a general target update can be approximated by one routed branch followed by one local branch.

Let $K \subseteq V^S$ be compact, let $\tilde{K} \subseteq V^S$ be a compact ambient set, and define

$$\Phi_\varepsilon(H) = H + \varepsilon \Delta(H).$$

Assume that on K ,

$$\Delta(H) = A(H) + L(H) + R_{\text{mix}}(H), \tag{1}$$

where A is the selected routed contribution, L the selected local contribution, and R_{mix} the residual structural defect. Let $\hat{A}$ and $\hat{L}$ be realizations satisfying

$$K \cup \{H + \varepsilon \hat{A}(H) : H \in K\} \subseteq \tilde{K}.$$

Define

$$\begin{aligned}\rho_\infty &:= \sup_{H \in K} \|A(H) - \hat{A}(H)\|, \\ m_\infty &:= \sup_{H \in K} \|R_{\text{mix}}(H)\|, \\ \delta_\infty &:= \sup_{H \in \tilde{K}} \|L(H) - \hat{L}(H)\|, \\ \hat{a} &:= \sup_{H \in K} \|\hat{A}(H)\|.\end{aligned}$$

Assume L is Λ -Lipschitz on $\tilde{K}$.

Theorem 1.4 (Generic staged approximation). *The sequential approximation*

$$H^{(1)} = H + \varepsilon \hat{A}(H), \quad T_\varepsilon(H) = H^{(1)} + \varepsilon \hat{L}(H^{(1)})$$

satisfies

$$\sup_{H \in K} \|\Phi_\varepsilon(H) - T_\varepsilon(H)\| \leq \varepsilon (m_\infty + \rho_\infty + \delta_\infty) + \varepsilon^2 \Lambda \hat{a}.$$

Proof. For $H \in K$, use (1) to write

$$\begin{aligned}\Phi_\varepsilon(H) - T_\varepsilon(H) &= \varepsilon(A(H) - \hat{A}(H)) + \varepsilon R_{\text{mix}}(H) \\ &\quad + \varepsilon \left[L(H) - \hat{L}(H + \varepsilon \hat{A}(H)) \right].\end{aligned}$$

Insert and subtract $L(H + \varepsilon \hat{A}(H))$. The first two terms are bounded by $\varepsilon(\rho_\infty + m_\infty)$. The local realization defect contributes at most $\varepsilon \delta_\infty$. Lipschitz continuity gives

$$\varepsilon \|L(H) - L(H + \varepsilon \hat{A}(H))\| \leq \varepsilon^2 \Lambda \|\hat{A}(H)\| \leq \varepsilon^2 \Lambda \hat{a}.$$

Take the supremum over K . □

The terms have different meanings:

- m_∞ measures failure of the selected routed-local structural hypothesis;
- ρ_∞ measures finite routed-realization error;
- δ_∞ measures finite local-realization error;
- $\varepsilon^2 \Lambda \hat{a}$ is the first schedule-composition correction.

No single term should be reported as “the Transformer approximation error” without declaring its source.

1.5 Controlled residual continuation

Residual addition supplies a marked identity path, but it does not unconditionally preserve every distinction after the branch is added.

Lemma 1.5 (Injective controlled residual). *Let $C \subseteq \mathbb{R}^d$, let $\phi : C \rightarrow \mathbb{R}^p$ be L_ϕ -Lipschitz, and let $J : \mathbb{R}^p \rightarrow \mathbb{R}^d$ be linear. Define*

$$\Psi_\varepsilon(z) = z + \varepsilon J\phi(z).$$

If

$$\varepsilon \|J\|_{\text{op}} L_\phi < 1,$$

then

$$\|\Psi_\varepsilon(z) - \Psi_\varepsilon(z')\| \geq \left(1 - \varepsilon \|J\|_{\text{op}} L_\phi\right) \|z - z'\|.$$

Hence Ψ_ε is injective on C and its inverse on $\Psi_\varepsilon(C)$ is Lipschitz with constant at most

$$\left(1 - \varepsilon \|J\|_{\text{op}} L_\phi\right)^{-1}.$$

Proof. By the reverse triangle inequality,

$$\begin{aligned}\|\Psi_\varepsilon(z) - \Psi_\varepsilon(z')\| &\geq \|z - z'\| - \varepsilon \|J\|_{\text{op}} \|\phi(z) - \phi(z')\| \\ &\geq \left(1 - \varepsilon \|J\|_{\text{op}} L_\phi\right) \|z - z'\|.\end{aligned}$$

The positive lower factor gives injectivity and the inverse Lipschitz estimate. $\square$

Without such a condition, an additive residual map can be noninjective. Identity continuation is a marked architectural path, not an unconditional theorem that the total successor map preserves all information.

1.6 Finite-support interpolation

Proposition 1.6 (Finite represented-state interpolation). *Let $z_1, \dots, z_M \in \mathbb{R}^d$ be distinct and let $y_1, \dots, y_M \in \mathbb{R}^p$. There exists a finite-width two-layer ReLU network f such that*

$$f(z_m) = y_m$$

for every m .

Proof. Choose $a \in \mathbb{R}^d$ outside the finite union of hyperplanes

$$\bigcup_{m \neq n} \{a : a^\top (z_m - z_n) = 0\}.$$

Then the scalars $t_m = a^\top z_m$ are pairwise distinct. Order them and construct a vector-valued continuous piecewise-linear map $g : \mathbb{R} \rightarrow \mathbb{R}^p$ satisfying $g(t_m) = y_m$. Every scalar continuous piecewise-linear function on the line is an affine function plus a finite sum of ReLU hinges; apply this coordinatewise to g . Then

$$f(z) := g(a^\top z)$$

is a finite-width two-layer ReLU network with the required interpolation values. $\square$

Finite-sample memorization therefore does not identify architecture. The relevant question is which marked interfaces, sharing laws, and continuation structure generated those values and how the construction behaves beyond the finite support.

2 Variants, edge cases, and qualification ledger

This section records boundary conditions that prevent the main results from being read more broadly than proved.

2.1 Masks and empty support

Receiver-row anchoring requires at least one admissible source. If a row has empty support, the architecture must declare a separate fallback, such as:

- identity continuation with zero routed increment;
- a fixed null source;
- a learned default carrier;
- an instance-supplied fallback state;
- an explicit error or halting state.

Softmax does not define the empty-row case by itself.

In the main attention normal form, the hard mask is fixed with respect to the represented state H on the declared domain. Its marked provenance may still be:

- schema-fixed, as in a causal order;
- instance-supplied, as in padding or a graph;
- parameter-dependent but state-independent.

The theorem permits any such declared mask with nonempty rows. A state-generated mask $M_{ri}(H)$ is a neighboring dynamic-support regime: the same row formula may be applied on each realized support, but support endogenization is then an additional architectural commitment rather than part of the fixed-mask theorem.

2.2 Temperature and hard routing

The attention normal form assumes a finite positive inverse-temperature scale β . As $\beta \rightarrow \infty$, the row distribution concentrates on score maximizers, but a tied maximum does not determine a unique hard route without an additional tie-breaking or set-valued rule. Hard top- k routing is therefore either:

- (i) a different receiver anchor;
- (ii) a separately declared discontinuous limit with a tie convention;
- (iii) a stochastic choice rule on the maximizing set.

It should not be treated as literally the same map as finite-temperature softmax.

2.3 Signed, vector, and operator-valued routing

The positive scalar evidence regime excludes cancellation at the routing-coefficient level. Signed scalar routing may sometimes be represented by expanding the carrier and moving sign into payload directions, as in the FFN signed-carrier construction. That expansion changes the marked presentation and does not establish identity between every signed operator and a positive simplex allocation.

Vector- or operator-valued routing has the form

$$z_r(H) = \sum_i T_{ri}(H) u_i(H),$$

with $T_{ri}(H)$ itself carrying channel- or coordinate-dependent action. Such transport cannot in general be reduced to one scalar coefficient per pair without expanding the carrier, increasing payload type, or losing information. Scalar attention is therefore one codomain choice for routed organization, not the universal routed form.

2.4 Finite rank and shared separability

Every realized finite score matrix has finite matrix rank. This pointwise fact does not supply shared query and key maps across a state domain. The relevant main-text assumption is family-level separability:

$$s_{ri}(H) = \langle q(x_r(H)), k(y_i(H)) \rangle$$

with shared receiver and source feature maps x_r, y_i and shared functions q, k . Any fixed positive pair bias is represented once through the baseline factor κ_{ri} rather than duplicated inside the contextual score. Independent factorizations chosen separately for each realized H may:

- vary discontinuously with H ;
- depend jointly on the entire state rather than on receiver and source states separately;
- require incompatible coordinate gauges across states;
- fail to define one shared parameterized query-key architecture.

The rank-truncation bound in Proposition 1.3 applies after one common bilinear score operator has been declared.

2.5 Source-local payloads

Standard attention uses a payload $v_i(y_i(H))$ that factors through the marked source feature and is independent of the receiver. A general source-resolved message may instead have

$$m_{ri}(H) = M(x_r(H), y_i(H), e_{ri}, H),$$

with receiver-, relation-, or global-state-conditioned content. Such a message remains a valid receiver presentation but lies outside the source-local-payload assumption of the attention normal form. This distinction is independent of whether routing coefficients are normalized.

2.6 Aggregation dimension and target sufficiency

The aggregate-fiber obstruction applies to the actual receiver exposure, whatever its dimension. Increasing width can refine exposure but does not guarantee injectivity on the relevant state domain. Conversely, a low-dimensional exposure may be sufficient for a low-dimensional target.

Two questions should be separated:

- (i) **architecture-relative**: which distinctions does the receiver interface retain?;
- (ii) **target-relative**: which of those distinctions are needed for the selected continuation or scientific query?

A representation may be sufficient for one target and structurally destructive for another.

2.7 Stochastic stages

The main article uses deterministic maps. A stochastic stage can be treated in at least two ways.

Explicit randomness. Enlarge the predecessor state by a random seed ω and analyze a deterministic map

$$F : X \times \Omega_{\text{rand}} \rightarrow Y.$$

Fiber claims then hold pointwise or almost surely on the enlarged state.

Markov-kernel presentation. Represent the stage by a kernel

$$K(x, \cdot) \in \mathcal{P}(Y).$$

Exact descent requires equality of conditional output laws on source-representation fibers. Receiver-interface factorization must then be formulated in a category of kernels or probability measures.

Training-time dropout is excluded from the canonical deterministic block theorem for this reason. Its inference-time deterministic scaling is a different stage.

2.8 Carrier changes and cross-attention

When source and receiver carriers differ, additive routing remains typed:

$$A_H : \bigoplus_{i \in S} U_i \rightarrow \bigoplus_{r \in R} W_r.$$

Cross-attention is therefore covered by distinct source and receiver carriers. Pooling, patch merging, carrier creation, deletion, or splitting additionally require an explicit successor carrier presentation. Identity residual continuation is unavailable when predecessor and successor local state types do not match, unless an embedding or typed transport is declared.

A carrier-changing map may be:

- injective expansion;

- invertible typed recoding;
- quotient or pooling;
- many-to-many transport through a new carrier;
- state-conditioned carrier generation.

Noninvertible carrier changes create new interface fibers and therefore additional information barriers.

2.9 Parallel, sequential, and recurrent schedules

A parallel block

$$H^+ = H + A(H) + L(H)$$

is not the same marked schedule as the sequential block

$$H^{(1)} = H + A(H), \quad H^+ = H^{(1)} + L(H^{(1)}).$$

The sequential local branch is evaluated on a different represented state. Even if both expressions have the same first-order expansion under a small step size, they define different exact stage factorizations and different intermediate interfaces.

Recurrent or equilibrium blocks introduce additional objects:

- a recurrent state or iterate carrier;
- a stopping or convergence condition;
- a fixed-point selection rule when solutions are nonunique;
- a schedule for exposing intermediate iterates.

These are architectural commitments rather than numerical details.

2.10 Residual continuation is not unconditional invertibility

An identity residual path marks predecessor state as one summand of the successor expression. The total map

$$z \mapsto z + f(z)$$

can nevertheless be noninjective. Lemma 1.5 gives one sufficient small-Lipschitz condition for injectivity. Absent such a condition, the correct claim is:

The architecture contains an identity continuation path in its marked factorization.

It is stronger, and generally false without hypotheses, to claim that the complete residual update preserves every predecessor distinction.

2.11 Architecture comparison

The main article proves explicit typed realization witnesses, such as the schema-fixed additive routed realization of finite convolution. It does not define a universal specialization preorder over arbitrary schemas. Such a preorder would require alignment maps between carrier and stage types, composition rules for fixing and endogenizing slots, treatment of schedule changes, and a declaration of which marks are constitutive for the comparison. The local re-presentation convention handles only bijective recodings of one receiver interface. The paper therefore uses explicit realization witnesses and coordinate profiles rather than asserting one total ordering of architectures.

2.12 Routing and stronger mediation claims

A routing coefficient belongs to the operative allocation rule. It is not automatically an intervention-relative edge. Distinct allocations over identical payloads can yield the same receiver aggregate, and aggregation or later continuation can cancel source variation. Establishing a stronger mediation claim requires independently declared admissible contrasts, controls, and receiver outcomes. No such relation follows from the attention formula alone.

2.13 Distinct architectural work performed by the attention commitments

The normal-form assumptions select different coordinates of the attention realization. The following witnesses remove one commitment while retaining a typed receiver factorization.

Nonmultiplicative positive link. Let $\psi(s) = \log(1 + e^s)$ and set

$$\alpha_{ri} = \frac{M_{ri} \kappa_{ri} \psi(s_{ri})}{\sum_{j \in S_r} \kappa_{rj} \psi(s_{rj})}.$$

The row remains positive, source-local, and additive, but $\psi(a + b) \neq \psi(a)\psi(b)$ in general. The exponential evidence law therefore does not follow.

Signed or operator-valued pair contributions. Let $C_{ri}(H) \in \mathcal{L}(U_i, W_r)$ and define

$$z_r(H) = \sum_i C_{ri}(H) u_i(H).$$

This satisfies the additive routed realization of the main article without a factorization into positive scalar mass and source-local payload. A probability row, scalar potential, and KL free-energy characterization are therefore unavailable without further marks.

Globally coupled allocation. Let $w_{ri} > 0$ and choose α by minimizing a transport objective subject to both row sums and column capacities. Ratios within one row then depend on constraints involving other receivers. Independent receiver-row anchoring and rowwise softmax no longer follow, even though the allocation remains positive and normalized.

Pair-conditioned payloads. Let

$$z_r(H) = \sum_i \alpha_{ri}(H) v(x_r(H), y_i(H)).$$

The coefficients may still be standard softmax coefficients, but the transported value no longer factors through the source feature alone. The standard source-only value map is therefore an additional architectural commitment.

Nonadditive aggregation. Let a receiver state evolve as

$$h_{r,0} = 0, \quad h_{r,t+1} = \Phi(h_{r,t}, c_{r,i_t}), \quad z_r = h_{r,|S_r|},$$

for a declared source order and nonlinear update Φ . Source-resolved contributions remain available, but one weighted-sum exposure does not describe what survives.

A nonseparable score family. On $[0, 1]^2$, the family

$$s(x, y) = e^{xy}$$

has no exact finite separation-rank representation on the full continuous domain. Indeed, for any distinct $x_1, \dots, x_n$, the functions $y \mapsto e^{x_m y}$ are linearly independent: differentiating a putative relation at $y = 0$ through order $n - 1$ gives a Vandermonde system. Hence the score family spans no finite-dimensional function space in either argument. Positive row anchoring still gives a normalized kernel, but no exact shared finite-dimensional query–key chart follows.

State-generated support. Let

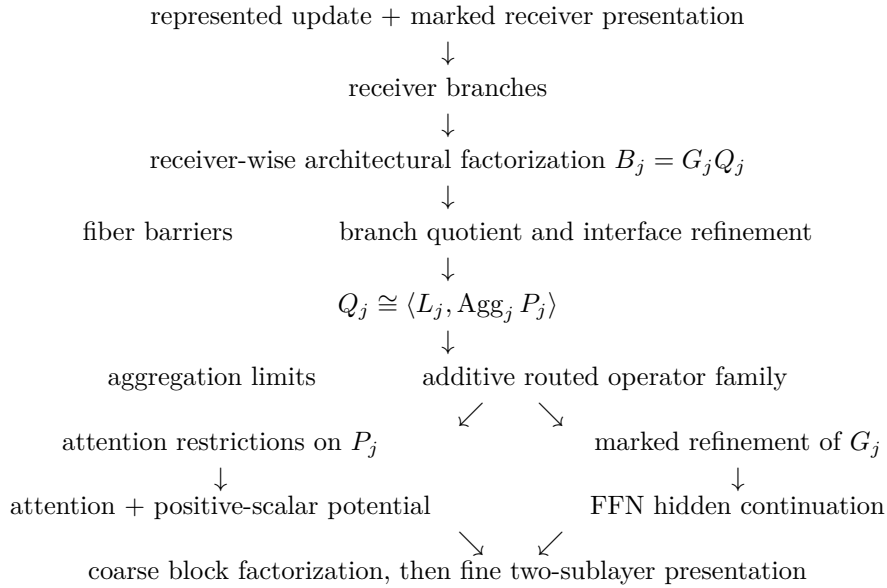
$$M_{ri}(H) = \mathbf{1}\{g(x_r(H), y_i(H)) \geq 0\}.$$

The allocation may remain positive, source-local, and additive on its realized support, but admissibility is now constructed from current state rather than supplied as a fixed hard sector. The resulting architecture has a different provenance coordinate even when a realized row numerically matches a fixed mask.

These examples do not claim that the listed assumptions are logically necessary for every use of the word “attention.” They show that each assumption performs distinct architectural work in the stated normal form.

2.14 Proof dependency audit

The main logical dependencies are:



The aggregate-fiber theorem branches directly from the source-resolved refinement and constrains every later local continuation. Provenance typing and essential dependence constrain interpretation across the diagram. Stochastic successor laws and other extensions in this supplement are not prerequisites for any arrow above.

2.15 Claim-status ledger

Table 1: Conditional results and explicitly rejected overclaims.

Claim	Status	Support
Representation fibers impose exact barriers	proved	fiber factorization criterion
Carrier indexing alone imposes locality	rejected	full-access collapse
Locality survives arbitrary global coordinate mixing	rejected	conjugacy counterexample
A proposed receiver interface is sufficient	iff theorem	fiber factorization criterion
Every re-presentation preserves receiver locality	rejected without matching interface fibers	conjugacy counterexample and re-presentation proposition
State-indexed additive routing defines an operator family	proved	additive routed realization
State conditioning uniquely implies attention	rejected	attention commitment ledger

Claim	Status	Support
Declared scalar positive commitments imply soft-max attention	proved	attention normal-form theorem
Pairwise compute equals retained exposure	rejected	aggregate-fiber obstruction
A later local map can recover erased distinctions	rejected	deterministic lower bound
Plain and gated FFNs admit hidden-direction routing	proved	exact column decompositions
A canonical PreNorm block factors through an attention-constructed interface and structured local continuation	proved	coarse whole-block theorem
The coarse or fine decomposition uniquely identifies mechanisms	rejected	factorization and gauge underdetermination
Attention weights are direct causal edges	rejected	routing/mediation distinction
Transformers are assumption-free or inevitable	rejected	remaining-commitment audit

3 Results moved from the main dependency spine

This section records exact results that remain useful but are not required to identify the minimal receiver-factorization object or derive the Transformer normal form.

3.1 Information lost under a deterministic aggregate

Proposition 3.1 (Evidence accounting under a deterministic quotient). *Let Y and Z be finite-valued random variables and let $Z' = q(Z)$ be deterministic. Then*

$$I(Y; Z) = I(Y; Z') + I(Y; Z \mid Z').$$

Equivalently,

$$H(Y \mid Z') - H(Y \mid Z) = I(Y; Z \mid Z') \geq 0.$$

Proof. Because Z' is a function of Z , the pair (Z, Z') contains the same information as Z . The chain rule gives

$$I(Y; Z, Z') = I(Y; Z') + I(Y; Z \mid Z'),$$

and the left side is $I(Y; Z)$. The entropy form follows from the definition of conditional mutual information. $\square$

Applied to a source-resolved presentation $P(H)$ and aggregate $E(H) = \text{Agg } P(H)$, the conditional term measures target-relevant evidence present before aggregation and absent afterward.

3.2 Signed and normalized presentations of an FFN

Let

$$\text{FFN}(x) - b_2 = \sum_e a_e(x) w_e.$$

Corollary 3.2 (Signed nonnegative presentation). *Define*

$$a_e^+ = \max\{a_e, 0\}, \quad a_e^- = \max\{-a_e, 0\},$$

and the doubled carrier $\tilde{E} = (\{+\} \times E) \sqcup (\{-\} \times E)$ with payloads $v(+, e) = w_e$ and $v(-, e) = -w_e$. Then

$$\text{FFN}(x) - b_2 = \sum_{(\sigma, e) \in \tilde{E}} a_e^\sigma(x) v(\sigma, e)$$

with nonnegative coefficients.

Proof. Use $a_e = a_e^+ - a_e^-$ and sum coordinatewise. □

Corollary 3.3 (Amplitude-allocation-direction form). *If*

$$A(x) := \sum_e (a_e^+(x) + a_e^-(x)) > 0,$$

define $p_x(\sigma, e) = a_e^\sigma(x)/A(x)$. Then p_x is a probability law and

$$\text{FFN}(x) - b_2 = A(x) \sum_{\sigma, e} p_x(\sigma, e) v(\sigma, e).$$

When $A(x) = 0$, the routed contribution vanishes.

This construction separates total amplitude from a normalized hidden-direction allocation. Defining $-\tau \log p_x$ gives a tautological hidden potential on the positive support, but not the compositional-evidence result obtained for attention.

3.3 Finite typed local approximation

Theorem 3.4 (Finite typed family approximation). *Let T be finite. For each $t \in T$, let $K_t \subset \mathbb{R}^d$ be compact and $f_t : K_t \rightarrow \mathbb{R}^p$ continuous. Let $e_t \in \mathbb{R}^{|T|}$ be one-hot. For every $\varepsilon > 0$, there exists one two-layer network $N : \mathbb{R}^{d+|T|} \rightarrow \mathbb{R}^p$ with continuous nonpolynomial activation such that*

$$\sup_{t \in T} \sup_{x \in K_t} \|N(x, e_t) - f_t(x)\| < \varepsilon.$$

Proof. The set $K = \bigcup_t (K_t \times \{e_t\})$ is compact. The function $f(x, e_t) = f_t(x)$ is continuous on this finite disjoint union. Universal approximation gives the result [Leshno et al., 1993]. □

Type marks can therefore condition one shared local continuation without being erased from the interface.

3.4 Two-stage residual error propagation

Let

$$U = H + T(H), \quad B(H) = U + F(U),$$

and let

$$\hat{U} = H + \hat{T}(H), \quad \hat{B}(H) = \hat{U} + \hat{F}(\hat{U}).$$

Proposition 3.5 (Two-stage residual error propagation). *Suppose*

$$\|T(H) - \hat{T}(H)\| \leq \varepsilon_T,$$

$$\sup_{V \in D} \|F(V) - \hat{F}(V)\| \leq \varepsilon_F,$$

and F is L_F -Lipschitz on a domain containing U and $\hat{U}$. Then

$$\boxed{\|B(H) - \hat{B}(H)\| \leq (1 + L_F)\varepsilon_T + \varepsilon_F.}$$

Proof. First $\|U - \hat{U}\| \leq \varepsilon_T$. Then

$$\begin{aligned} \|B(H) - \hat{B}(H)\| &\leq \|U - \hat{U}\| + \|F(U) - \hat{F}(\hat{U})\| \\ &\leq \varepsilon_T + L_F \varepsilon_T + \varepsilon_F. \end{aligned}$$

□

The ledger separates approximation error in the routed interface from approximation error in the local continuation. It begins after the exact interface has been chosen and therefore does not remove an interface-fiber lower bound.

4 Stochastic variable-successor realizations

The main article introduces deterministic updates into a tagged union $Y^\sqcup = \bigsqcup_{c \in \mathfrak{C}} Y_c$. This section records a stochastic realization. The stochastic structure is an extension of the architectural target, not a prerequisite for the deterministic Transformer reconstruction.

4.1 Configuration and within-sector content

Equip each Y_c and the disjoint union $Y^\sqcup$ with measurable structures. A stochastic successor update is a Markov kernel

$$K_\theta : X \rightsquigarrow Y^\sqcup.$$

Let $\Pi_\theta(\cdot | H)$ be a probability law on $\mathfrak{C}$, and for each c let $Q_{\theta,c}(\cdot | H)$ be a probability kernel supported on Y_c . For measurable $A \subseteq Y^\sqcup$, define

$$K_\theta(A | H) = \sum_{c \in \mathfrak{C}} \Pi_\theta(c | H) Q_{\theta,c}(A \cap Y_c | H). \quad (2)$$

The sector law selects the successor receiver presentation; the within-sector kernel selects content inside it. A deterministic update is the degenerate case in which both kernels are point masses.

4.2 Positive joint successor law

Let ρ be a probability reference measure on $Y^\sqcup$, let $\mathcal{U}_\theta(x^+; H)$ be a measurable contextual successor potential, and let $C(x^+)$ be a measurable successor penalty. Assume

$$0 < \Xi_\theta(H) := \int_{Y^\sqcup} \exp[-(\mathcal{U}_\theta(x; H) + C(x))/\tau] \rho(dx) < \infty.$$

Define

$$\mathbb{P}_\theta(dx^+ | H) = \frac{\exp[-(\mathcal{U}_\theta(x^+; H) + C(x^+))/\tau] \rho(dx^+)}{\Xi_\theta(H)}. \quad (3)$$

The term C is a declared penalty, not automatically a physical or computational cost. When it factors through sect, it is a configuration penalty. Carrier count alone need not equal FLOPs, latency, storage, energy, or description length.

Proposition 4.1 (Effective sector potential). *Let ρ_c be the restriction of ρ to Y_c and define*

$$Z_c(H) := \int_{Y_c} \exp[-(\mathcal{U}_\theta(x; H) + C(x))/\tau] \rho_c(dx).$$

Set $\mathcal{U}_\theta^{\text{eff}}(c; H) := -\tau \log Z_c(H)$ when $Z_c(H) > 0$, and $+\infty$ otherwise. The sector marginal of (3) is

$$\Pi_\theta^{\text{Gibbs}}(c | H) = \frac{e^{-\mathcal{U}_\theta^{\text{eff}}(c; H)/\tau}}{\sum_{c'} e^{-\mathcal{U}_\theta^{\text{eff}}(c'; H)/\tau}}.$$

Proof. Integrate the joint numerator over each disjoint sector. The sector integrals partition the global normalizer. $\square$

Integrating out within-sector content therefore gives a log-integral-exp potential over configurations. This is a stochastic quotient identity, not evidence that the architecture is physically equilibrated.

4.3 Joint successor free energy

Let $\mathcal{E}(\cdot; H) = \mathcal{U}_\theta(\cdot; H) + C(\cdot)$. For every probability law $p \ll \rho$ for which $\mathbb{E}_p[|\mathcal{E}(X; H)|]$ and $D_{\text{KL}}(p || \rho)$ are finite, define

$$\mathcal{G}[p; H] := \mathbb{E}_p[\mathcal{E}(X; H)] + \tau D_{\text{KL}}(p || \rho).$$

Theorem 4.2 (Joint successor free-energy characterization). *For the law in (3),*

$$\mathcal{G}[p; H] = -\tau \log \Xi_\theta(H) + \tau D_{\text{KL}}(p \| \mathbb{P}_\theta(\cdot | H)).$$

Consequently, $\mathbb{P}_\theta(\cdot | H)$ is the unique minimizer among the admissible laws whenever its free energy is finite.

Proof. Use

$$\log \frac{d\mathbb{P}_\theta}{d\rho}(x) = -\frac{\mathcal{E}(x; H)}{\tau} - \log \Xi_\theta(H)$$

inside $D_{\text{KL}}(p \| \mathbb{P}_\theta)$ and rearrange. Nonnegativity of KL divergence gives minimality, with equality only when $p = \mathbb{P}_\theta(\cdot | H)$ almost everywhere. $\square$

4.4 Multi-token specialization and nonclaims

For language generation, take

$$Y_N = \Sigma^N, \quad Y^\sqcup = \bigsqcup_{N=0}^K \Sigma^N.$$

The sector variable is the proposed count N , while the within-sector law describes a jointly structured token block. Existing multi-token prediction and speculative-decoding methods are nearby realization strategies, but the construction above does not assert equality with them. It supplies no tractable partition function, efficient sampler, acceptance rule, equality with the original autoregressive law, or guaranteed compute reduction.

References

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